San Francisco Business Openings and Closings Pre- and Post-Pandemic

Urban Informatics Semester Project Midterm Report

Track B; MTC Team

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Repo link: https://github.com/miaflynn/CYPLAN255-Final-Project

Raw data: https://drive.google.com/file/d/1hQ6NLs8z3LlUk0f61ptz9fdgMmrP9K73/view?usp=sharing

Cleaned data: https://drive.google.com/drive/folders/1wo6Q6Y4GDdNumQ4DGk2WF9nmEfHL1whK?usp=sharing

1. Problem & motivation

Updated research question(s) with key terms clearly defined

Original: *How have the locations of registered businesses in San Francisco changed pre- and post-Pandemic? Are there any implications for Equity Priority Communities across the City?*

Revised:

- **Overarching Question:** How have the rates of openings, closures, and net change of San Francisco businesses varied across neighborhoods and business types pre- and post-pandemic (2016-2025), and to what extent do these patterns reveal spatial inequities in Equity Priority Communities?
- **Sub Questions:**
    - Which neighborhoods (and census tracts) experienced the highest rates of business closures during 2020-2021, and which show the most new openings (recovery) from 2022 to 2025?
    - Did Equity Priority Communities (EPCs) experience higher rates of business closures than non-EPC areas between 2016 and 2025? How does the rate of post-pandemic business openings in EPCs compare to non-EPCs?
    - Which types of businesses experienced higher rates of closures during the pandemic (2020-2021) and which types of businesses more strongly recovered post-pandemic (2022-2025)? How do these overall rates compare to Retail, Dining, and Entertainment (RDE) business types?

Key Terms

- **Registered Business:** Businesses that pay taxes to the City and County of San Francisco (DataSF)
- **Commercial revitalization:** Streamlining decision-making to facilitate real estate and public realm projects, assisting small businesses, attracting new employers, and financing workforce housing (SPUR).
- **Equity Priority Communities** (EPCs): MTC-designated census tracts that have a significant concentration of underserved populations, such as households with low incomes and people of color (MTC).
- **Retail, Dining, and Entertainment (RDE)**: A term used to describe businesses in the leisure sector centered around shopping, food, and recreation (Law Insider).

Stakeholders & Impact

San Francisco's commercial districts faced economic hardship during the COVID pandemic, causing many businesses to close. The pandemic created the largest disruption to the social and economic health of downtown SF since the 1906 and 1989 earthquakes (SPUR). While much has been written about the economic recovery of SF, few have discussed the spatial equity implications of the loss of local businesses and the lack of access to particular kinds of stores.

Sales tax revenue has recovered more in some neighborhoods than others, and we aim to investigate which neighborhoods were hit the hardest by different kinds of business closures and where economic resilience is concentrated (SF Chronicle).

Beyond the economic implications, such as loss of jobs and revenue, small, local, independent businesses in the City serve as cultural and community hubs, supporting the social fabric and daily life of diverse communities across the city. Understanding the spatial distribution of business recovery post-pandemic may ultimately help local governments understand which areas could benefit from targeted support for grants, loans, legal, and accounting services to prevent loss of local business and support recovery, especially for Equity Priority Communities, which have a higher proportion of low-income residents and people of color.

Unit of Analysis

- Census tract (to compare across neighborhoods and align with EPC boundaries)
- Point-level business locations for mapping change

Justification: There are tons of points in this dataset, so census tracts will allow for an aggregation of business changes and align with EPC designations, which are defined at the tract level. We will also look at point-level business locations for visualizing change across the city more broadly (and across types of businesses).

Measures & Metrics

- Closure rate = (number of businesses closed in timeframe) / (total businesses pre-timeframe)
- Opening rate = (# new businesses) / (number baseline businesses)
- Net business change = openings – closures
- Business density = businesses per tract/neighborhood
- EPC comparison = average values in EPC vs. non-EPC

2. Data & cleaning

SF Open Data Portal - Registered Business Locations *(The Treasurer & Tax Collector's Office collects data on business registration applications, account update/closure forms, and taxpayer filings, which are compiled into a dataset of businesses with start dates and end dates)*

- Coverage: Businesses registered within the City and County of SF; temporal coverage is 1849–2026 (however, businesses closed before 2014 have been deleted from the dataset).
- Granularity:
    - Spatial Granularity: point-level - street-level locations of businesses in San Francisco
    - Temporal Granularity: updated as business license changes - updated automatically (daily) from license portal.
- Type: Numeric and string/categorical data (coordinates, addresses, NAICS code, etc.)

- Format: Tabular (csv)
- Source: SF Open Data Portal (Official Government)

MTC Equity Priority Communities

- Coverage: Census tracts across SF Bay Area
- Granularity:
    - Spatial Granularity: Polygon at census tract level
    - Temporal Granularity: ACS Census 2014-2018 data
- Type: Vector geospatial data
- Format: Feature layer/Shapefile
- Source: MTC Open Data Portal

Cleaning process

- Did basic cleaning and reformatting of csv file, including setting crs, changing column names, and correcting datetime.
- Some businesses lacked a NAICS number to determine the type of business, so we filtered by businesses that had an NAICS code so we could code by business type.
    - Saved dataframe with all business types, made new dataframe containing only the following business types: Retail Trade; Food Services; Arts, Entertainment, and Recreation
- When mapping, we saw that businesses had geometries all over the country. This is because headquarters may be outside of the Bay Area, so we filtered businesses where the location existed within the boundaries of SF.
- Additionally, closings before the year 2014 were not documented, so we had to limit our timeframe of analysis to reduce error (post 2014, started our analysis in 2016).
- We filtered the data to remove the missing or errant data points listed above. We used valuecounts and plot to sanity check our data for inconsistencies.
- We verified/performed sanity checks of yearly trends for data that had outliers, compared counts before and after filtering (ex. NAICS and San Francisco boundaries), mapped raw vs. filtered data to confirm geography, and checked to confirm if outlier tracts with high #s to were real or due to data issues (ex. duplicates or address inconsistencies).
- No duplicates existed in the data.

3. Missing data analysis

Data We Know Are Missing: When the City transitioned to a new system in 2014, only active business accounts were migrated. As a result, any businesses that had already closed by that point were not included in the current dataset. Also, in 2018, the City did a cleanup of dormant and unresponsive accounts and closed approximately 40,000 inactive businesses.

Data We Know Are Missing: As described in the cleaning above, NAICS codes for some businesses are missing. We filtered by only businesses with this code bc we want to look at business types.

Data We Don't Know Are Missing: Businesses are marked administratively closed after not filing or communicating with the Treasurer and Tax Collector's Office for three years. Because we are working with data from within the last three years, there are likely businesses that have closed but not notified the office.

Missing What Matters*:* We are also missing data for informal businesses, such as food stalls and trucks, street vendors, and other forms of transient business.

Choosing Just Some Cases: We have decided to restrict our data timeline to eliminate missing data from the earlier years of recorded data. We have also decided to not consider informal businesses within our analysis, instead focusing on those housed within buildings to represent the permanent economic activities of each area (but we recognize this as a limitation of capturing the full picture).

4. Initial results & visualizations

Initial Findings

Overall Trends

- Total number of business openings and closings were increasing year over year until 2020. After 2020 (COVID pandemic), the total number of businesses closed and opened began decreasing year over year. After 2024, the number of businesses closed was higher than the number opened.

Trends by Census Tract

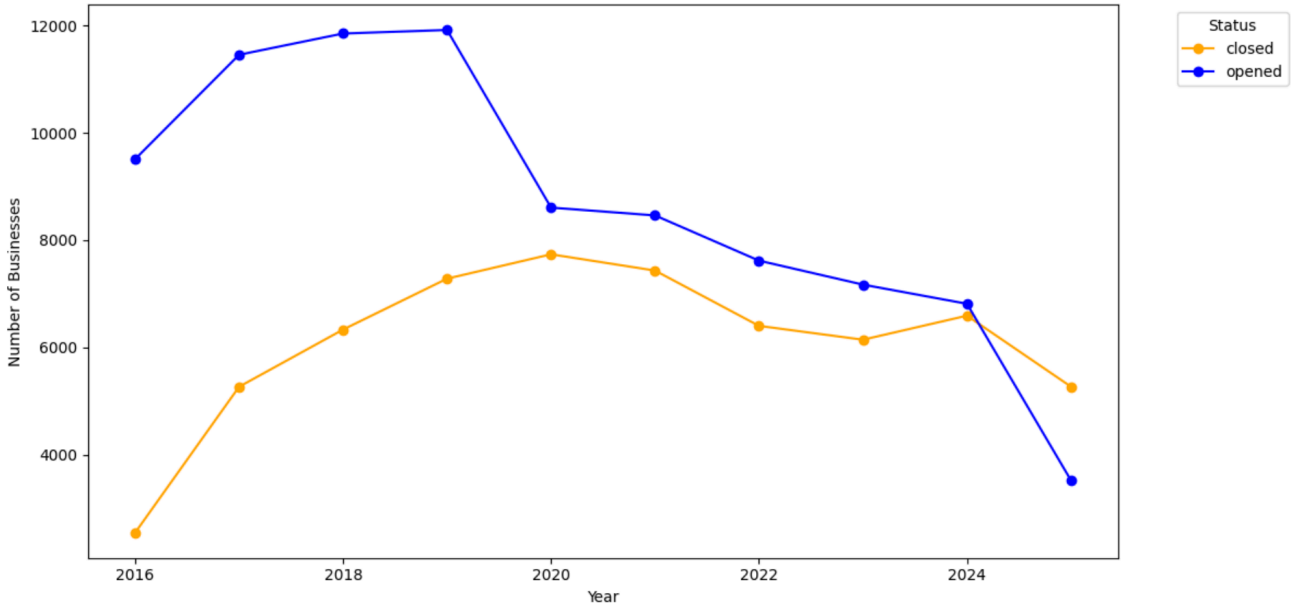
- Most census tracts show modest/little changes in business openings and closings compared to 2016 (median is +10 closings and -3 openings compared to 2016).
- A few tracts in downtown SF show extreme changes (ex. 400+ openings or closings in some years).

Trends for RDE Businesses

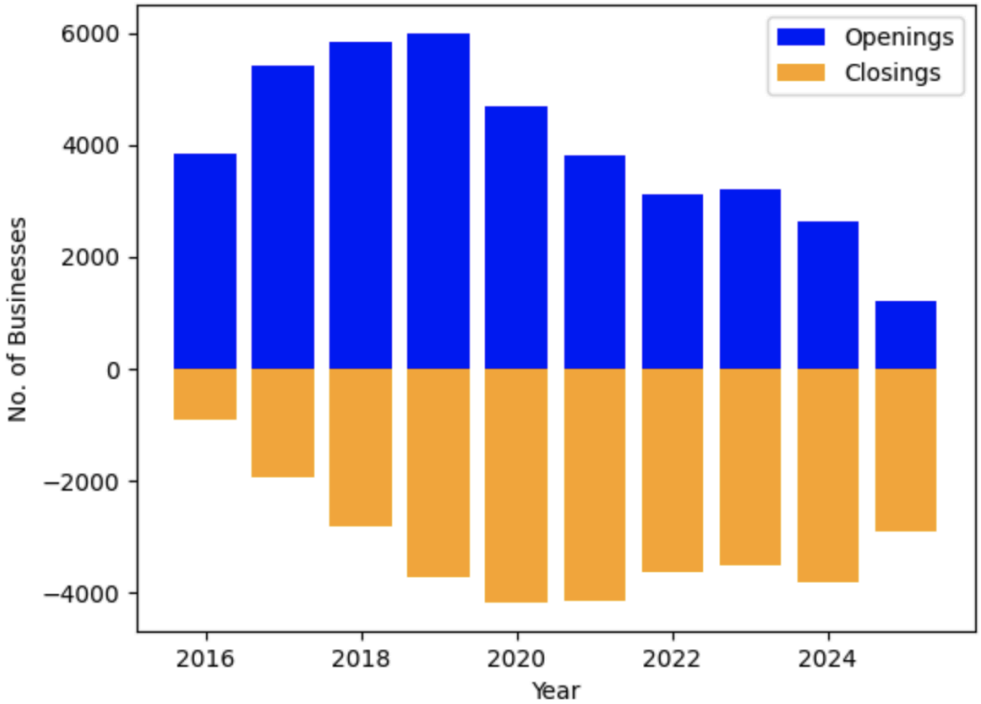
- Businesses in the recreation, dining, and entertainment industries faced significant closures during and after 2020, likely as a result of lack of business during the pandemic. These businesses have seen even fewer new openings post-pandemic, with the number of new businesses opening per year steadily falling across the timeframe.
- After 2020, closings of RDE businesses outpaced openings, a gap which continues to increase as time goes on.
- The yearly net declines are not evenly distributed across census tracts.

Charts and Maps

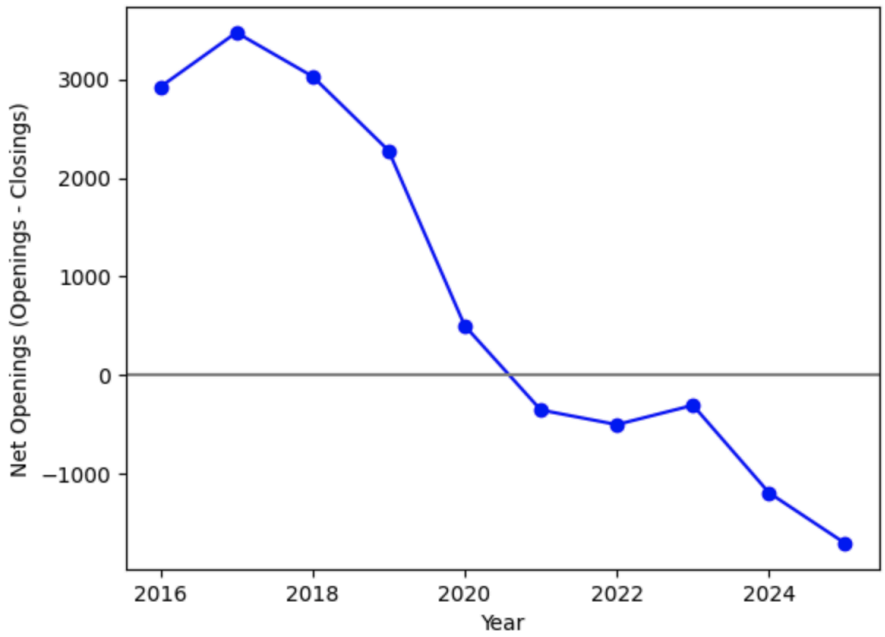
San Francisco Yearly Business Openings vs Closings (2016-2025)



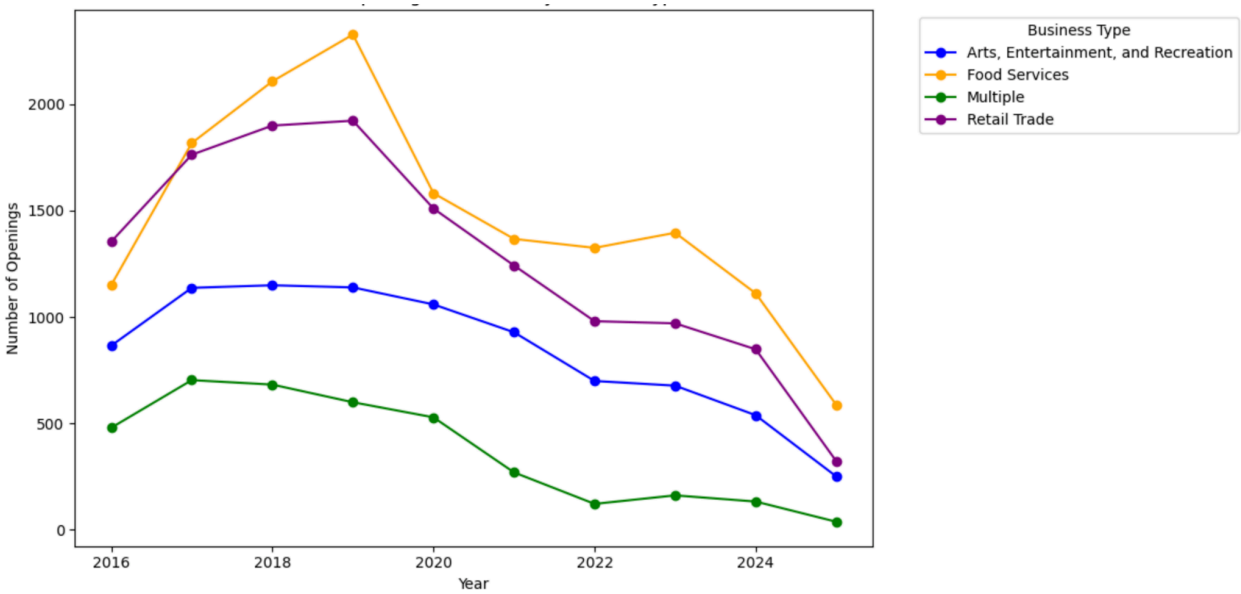
San Francisco Yearly RDE Openings and Closings (2016-2025)



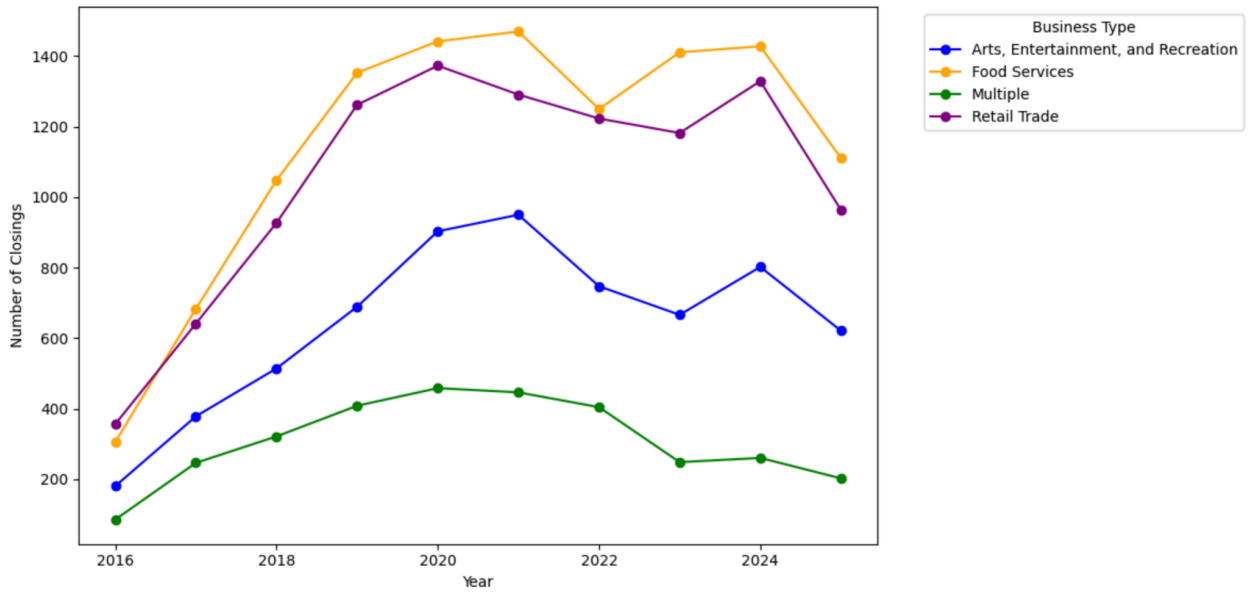
San Francisco Net RDE Business Quantity Change (2016-2025)



San Francisco Yearly RDE Business Openings by Category (2016-2025)



San Francisco Yearly RDE Business Closings by Category (2016-2025)

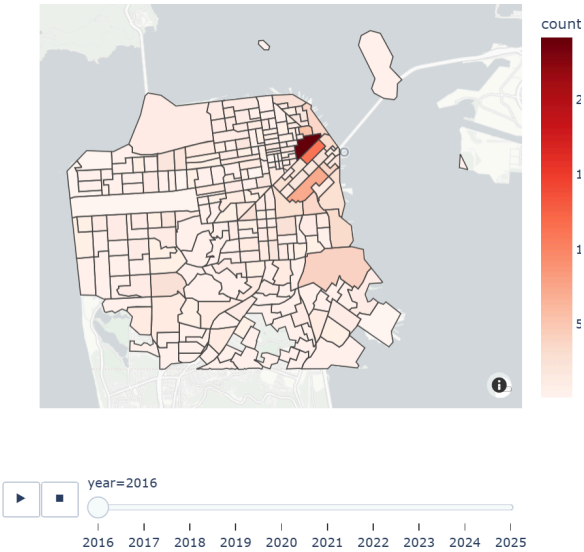


Openings and Closings by Census Tract

Screenshots of interactive maps below; example HTML: [SF Business Closings]

1. **SF Business Closings by Census Tract** (2016-2025) - *not yet baselined*
2. **SF Business Openings by Census Tract** (2016-2025) - *not yet baselined*
3. **SF Business Closings Change from 2016 by Census Tract** (2016-2025) - *2016 baseline*
4. **SF Business Openings Change from 2016 by Census Tract** (2016-2025) - *2016 baseline*

1



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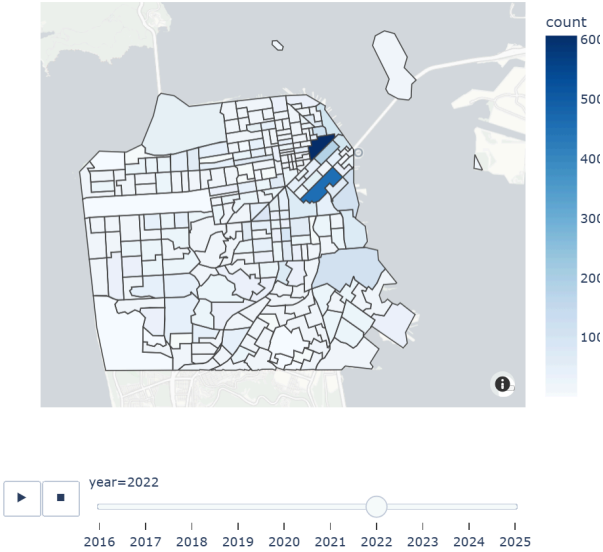

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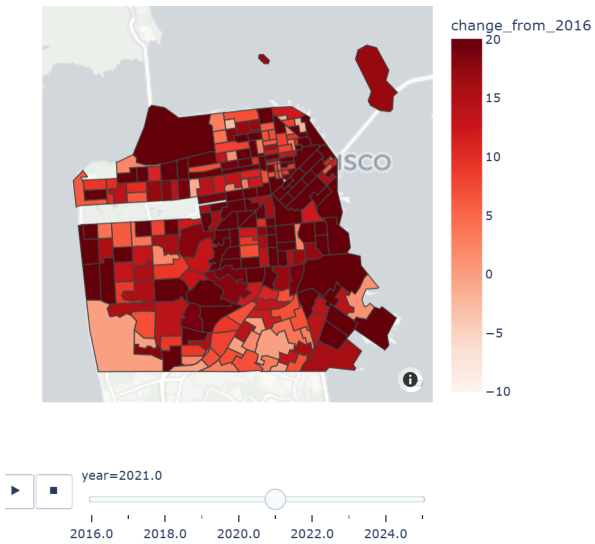

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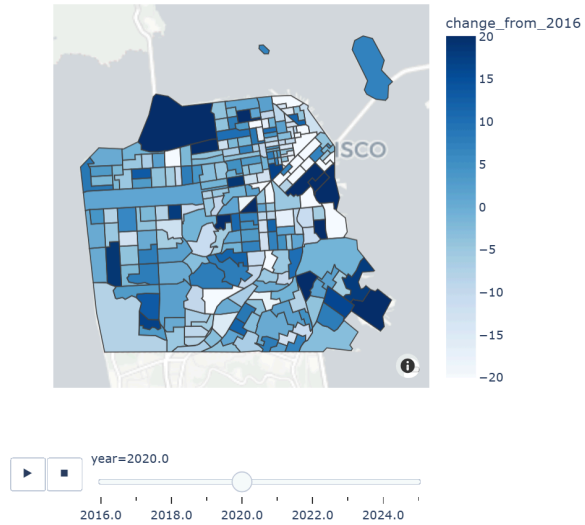
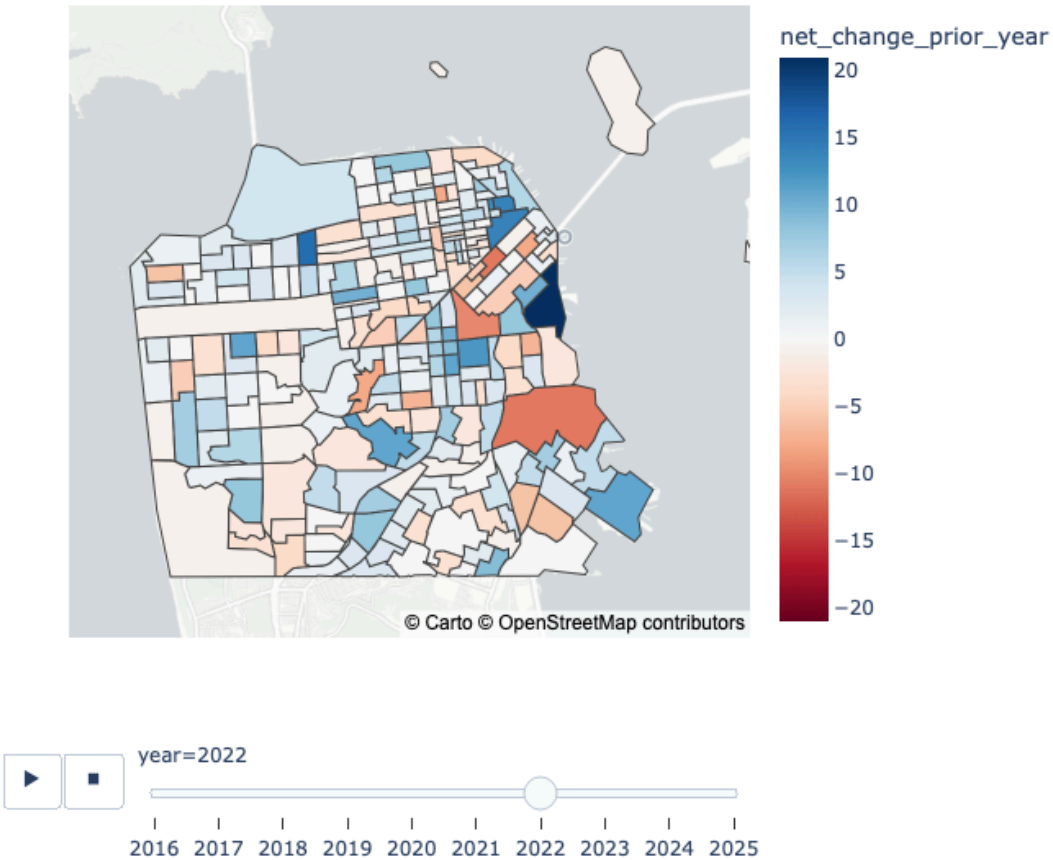


Notes/Caveats:

- For the midterm, we compared counts for each year to a 2016 baseline. However, we may choose a different baseline year (pre- and post-pandemic) or per capita in future visualizations.
- We also plan to aggregate by neighborhood instead of census tract to make it easier to see the changes on the map (and at a scale that people can better comprehend).
- Most tracts saw minimal closings/openings per year, but a few had 400+ closings and openings per year. To better visualize small changes, we made the color scale min and max smaller - will think through a better way to handle these outliers for the final.

San Francisco Net Change in RDE Businesses by Census Tract (yearly; 2016-2025)

Screenshot example of interactive visualization - will provide an interactive tool for final.



Notes/Caveats:

- The time series of net change each year shows that all tracts were consistently growing in RDE until 2020, when neighborhoods began to see net declines. From that point, the net change in RDE businesses is much more spatially varied, with growth occurring most in the tracts within the Financial District and Mission Bay.
- For the midterm, we looked at the net change each year from 2016 to 2025. We may choose a different year to start: either pre- and post-pandemic (2019-2025) or further back before 2016, in order to see the growth trends in greater context.
- We plan to aggregate my neighborhood to see if there are broader trends that tell a more understandable growth story.

5. Progress since proposal & plan to completion

Since our initial proposal, we have further refined our research question to focus on clear outcomes and measurable changes. We have also cleaned the data, begun analysis, and visualization.

From this point forward, we will primarily focus on visualization strategies and compare our findings, focusing on the general changes in business openings and closures over time, as well as the differences in business resiliency by neighborhood type.

From now until the submission, we are looking forward to regularly meeting with MTC to obtain guidance. We will spend the next 2-3 weeks working on charts/graphs highlighting key points in the data, as well as creating and refining an interactive tool to see changes over time. We will use the final weeks to make final changes, present our findings in an easy-to-access format, and craft our final report.

Timeline to completion:

- 4/1: Meeting with MTC (biweekly meetings following)
- 4/8: All analysis and data transformations done; workbook is clean and ready for visualization.
- 4/17: All visualization drafts done.
- 4/30: All reporting done and git hub updated; visualizations further refined.

6. References

Li, R., & Echeverria, D. (2025, October 17). *One S.F. neighborhood is seeing an economic boom. This map shows how bad things are in other areas*. San Francisco Chronicle. https://www.sfchronicle.com/sf/article/retail-economy-san-francisco-21102655.php

SPUR. (2024, April). *Small and mighty: How small businesses can reinvent downtown San Francisco* (Policy brief). SPUR. https://www.spur.org/sites/default/files/2024-05/SPUR_Small_and_Mighty_0.pdf